2.3.17

EE25BTECH11014 - Bhoomika Lokesh

Question: Find the properties of the triangle with vertices $(-4, 0)$,$(0, 3)$ and $(4, 0)$.
Solution:

Name	**Point**
A	$\begin{pmatrix} -4 \\ 0 \end{pmatrix}$
B	$\begin{pmatrix} 0 \\ 3 \end{pmatrix}$
C	$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

TABLE 0: Variables Used

Vectors of sides of the triangle are,

$$(\mathbf{A} - \mathbf{B}) = \begin{pmatrix} -4 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} -4 \\ -3 \end{pmatrix} \tag{0.1}$$

$$(\mathbf{B} - \mathbf{C}) = \begin{pmatrix} 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \end{pmatrix} \tag{0.2}$$

$$(\mathbf{A} - \mathbf{C}) = \begin{pmatrix} -4 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -8 \\ 0 \end{pmatrix} \tag{0.3}$$

To check for orthogonality, the dot product of the vectors of the sides of the triangle should be equal to zero.

$$(\mathbf{A} - \mathbf{B})^T \cdot (\mathbf{B} - \mathbf{C}) = \begin{pmatrix} -4 & -3 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 3 \end{pmatrix} = 16 - 9 = 7 \neq 0 \tag{0.4}$$

$$(\mathbf{B} - \mathbf{C})^T \cdot (\mathbf{A} - \mathbf{C}) = \begin{pmatrix} -4 & 3 \end{pmatrix} \cdot \begin{pmatrix} -8 \\ 0 \end{pmatrix} = 32 + 0 = 32 \neq 0 \tag{0.5}$$

$$(\mathbf{A} - \mathbf{C})^T \cdot (\mathbf{A} - \mathbf{B}) = \begin{pmatrix} -8 & 0 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ -3 \end{pmatrix} = 32 - 0 = 32 \neq 0 \tag{0.6}$$

Hence, the given triangle is **NOT RIGHT ANGLED**.

To check if the given triangle is isosceles,a median should be perpendicular to a side of the triangle.

Let the midpoints of the sides **AB**,**BC**,**b**e **x**, **y**and **z**.

$$\mathbf{x} = \frac{\mathbf{A}+\mathbf{B}}{2} = \begin{pmatrix}-4\\0\end{pmatrix} + \begin{pmatrix}0\\3\end{pmatrix} = \begin{pmatrix}-2\\\frac{3}{2}\end{pmatrix} \tag{0.7}$$

$$\mathbf{y} = \frac{\mathbf{B}+\mathbf{C}}{2} = \begin{pmatrix}0\\3\end{pmatrix} + \begin{pmatrix}4\\0\end{pmatrix} = \begin{pmatrix}2\\\frac{3}{2}\end{pmatrix} \tag{0.8}$$

$$\mathbf{z} = \frac{\mathbf{A}+\mathbf{C}}{2} = \begin{pmatrix}-4\\0\end{pmatrix} + \begin{pmatrix}4\\0\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix} \tag{0.9}$$

The vectors of medians of the triangle are:

$$\mathbf{x}-\mathbf{C} = \begin{pmatrix}-2\\\frac{3}{2}\end{pmatrix} - \begin{pmatrix}4\\0\end{pmatrix} = \begin{pmatrix}-6\\\frac{3}{2}\end{pmatrix} \tag{0.10}$$

$$\mathbf{y}-\mathbf{A} = \begin{pmatrix}2\\\frac{3}{2}\end{pmatrix} - \begin{pmatrix}-4\\0\end{pmatrix} = \begin{pmatrix}6\\\frac{3}{2}\end{pmatrix} \tag{0.11}$$

$$\mathbf{z}-\mathbf{B} = \begin{pmatrix}0\\0\end{pmatrix} - \begin{pmatrix}0\\3\end{pmatrix} = \begin{pmatrix}0\\\frac{3}{2}\end{pmatrix} \tag{0.12}$$

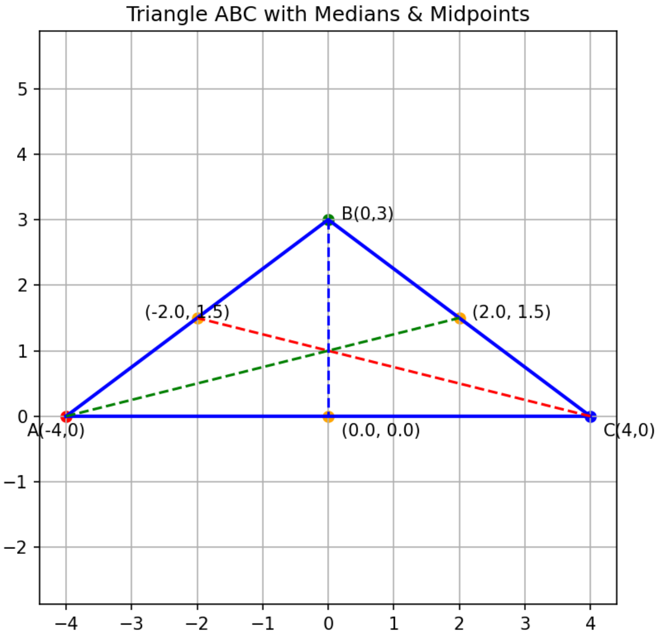
$$(\mathbf{x}-\mathbf{C})^T \cdot (\mathbf{A}-\mathbf{B}) = \begin{pmatrix}-6 & \frac{3}{2}\end{pmatrix} \cdot \begin{pmatrix}-4\\-3\end{pmatrix} = 24 - \frac{9}{2} = \frac{39}{2} \neq 0 \tag{0.13}$$

$$(\mathbf{y}-\mathbf{A})^T \cdot (\mathbf{B}-\mathbf{C}) = \begin{pmatrix}6 & \frac{3}{2}\end{pmatrix} \cdot \begin{pmatrix}-4\\3\end{pmatrix} = -24 + \frac{9}{2} = \frac{-39}{2} \neq 0 \tag{0.14}$$

$$(\mathbf{z}-\mathbf{B})^T \cdot (\mathbf{A}-\mathbf{C}) = \begin{pmatrix}0 & \frac{3}{2}\end{pmatrix} \cdot \begin{pmatrix}-8\\0\end{pmatrix} = 0 \tag{0.15}$$

$\therefore$ We can conclude that triangle **ABC** is an **ISOSCELES** triangle.

See Fig. 0.1



Triangle ABC with Medians & Midpoints
B(0,3)
(-2.0, 1.5)
(2.0, 1.5)
A(-4,0)
(0.0, 0.0)
C(4,0)
5
4
3
2
1
0
−1
−2
−4
−3
−2
−1
0
1
2
3
4

Fig. 0.1: Graph